

a simple step. Let's see how to do that:

- 1) Reboot your system and select the recovery mode from the GRUB menu (start-up menu). It would be something like 'Ubuntu 12.04 with Linux ... (recovery mode)'.
- 2) Drop to the root shell.
- 3) Give the command 'fsck' — this is the program that will help us in this situation. If it fails, try 'fsck help' or '*man fsck*' to find some solutions that will help you.



Note: `fsck` can also be run using a recovery disk (i.e., a live CD). However, running `fsck` with partitions that are mounted is a dangerous task.

Repairing GRUB

Many problems related to the OS can be solved by re-installing the bootloader. One day, I installed an old version of gNewSense and my Ubuntu disappeared. The reason was, gNewSense installed an older version of GRUB as the bootloader and it was unable to find the latest Ubuntu. I re-installed GRUB with a newer version and the problem was solved. The same problem can arise when you dual-boot with Microsoft Windows (that is why it is always recommended to install GNU/Linux after Windows). Such problems can be solved by the following method. Let's re-install the bootloader using a recovery disk (we are considering GRUB only since it is the most popular option):

- 1) Boot from the recovery disk that contains the latest version of GRUB.
- 2) Open a terminal and give the following command:

```
sudo grub-install <your-hard-disk>
```

Example:

```
sudo grub-install /dev/sda
```

As shown in the example, `/dev/sda` will be the hard disk in normal cases. You may use a disk utility to ensure that.

The process is now finished!

Additional information: The configuration file of GRUB is `/boot/grub/grub.cfg`. You can edit it manually — but be careful!

Preserving the 'Home' directory

'Home' is the most important directory for typical GNU/Linux users, since all their personal files are kept in it. It is a wonderful fact that a single Home directory can be shared between multiple GNU/Linux instances on a computer. This technique can be used to preserve your current Home directory when you install a new GNU/Linux distro. But don't forget to take a back-up before that.

Know more about the Home directory

The Home directory is where all the personal files of the users are kept. Your home directory's path is `/home/<username>`

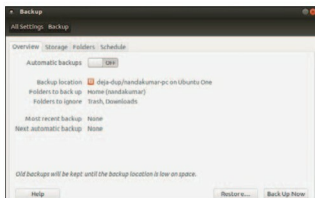


Figure 3: Home screen of TestDisk

(for example, my home directory is `/home/nandakumar`). `/home` can be set inside the `/` partition or as a new partition, based on your choice. The new partition is the best, since only this option allows you to share the Home. If you opt for the `/` partition, you have to manually copy all the contents of Home when you install a new GNU/Linux instance. Also, if the file system is damaged, you can only do a manual file recovery.

Note that the superuser's home directory is `/root`, which means it can't be shared/preserved using this sharing technique (it won't be necessary in most cases).



Note: Sharing the Home directory with different distros and different versions may cause instabilities. For example, sharing Home with Ubuntu 10.04 and Ubuntu 12.04 may make the Rhythmbox Playlist unusable. It'll hurt the themes, too. If you are upgrading from 10.04 to 12.04 by rewriting the / of 10.04 with 12.04 and preserving the Home, the sharing will be okay.

Sharing the Home directory between multiple GNU/Linux instances: You can share Home between multiple

GNU/Linux instances that are used by you frequently. But check its safety as mentioned earlier.

The labels and messages may vary with distros, but a general outline of the process is as follows:

- 1) Note the partition numbers of / and /home of the current OS and take a back-up of your Home directory.
- 2) Boot the installation medium of the new OS.
- 3) If you find something like 'Import user accounts and files from <your old OS>' during the installation, the best option would be to preserve your personal files (ensure that the files are also imported with user accounts). If so, you don't have to perform the upcoming steps. Else, select the option 'Manual partitioning', 'Advanced installation' or something like that under 'Where to install your OS?'.
 - 4) Then comes the partitioning dialogue box. Select a free space and choose the option 'Create'. Pick the size and set the mount point ('Use as') as /.
 - 5) Select the Home partition of your already-installed GNU/Linux instance. Edit it and set the mount point as /